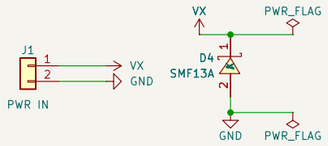
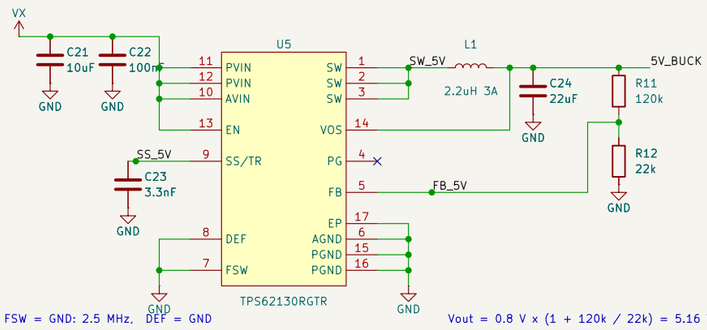
INPUT J1



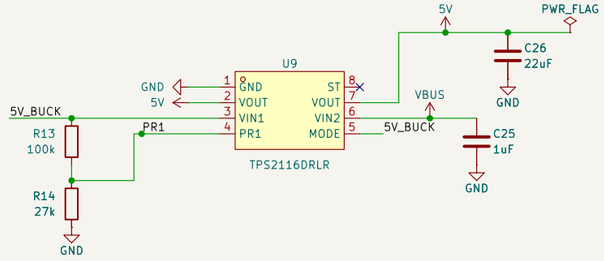
5V BUCK TPS62130 3 A, 2.5 MHz + VX input TVS



FSW = GND: 2.5 MHz, DEF = GND

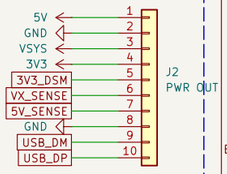
Vout = 0.8 V x (1 + 120k / 22k) = 5.16 V

5V PRIORITY MUX TPS2116: buck (VIN1) first, USB VBUS (VIN2) second

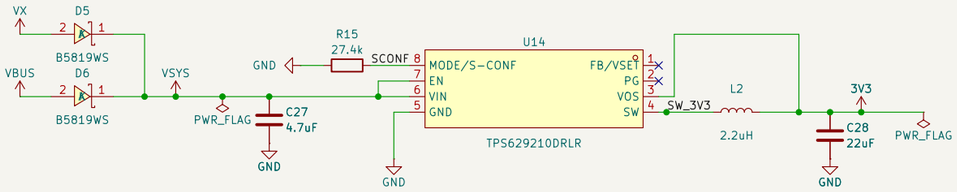


switches to USB below 1.0 V x (100k + 27k) / 27k = 4.7 V

OUTPUT J2



3V3 MCU BUCK TPS629210 1 A, fed from VSYS (VX or VBUS through Schottky OR)

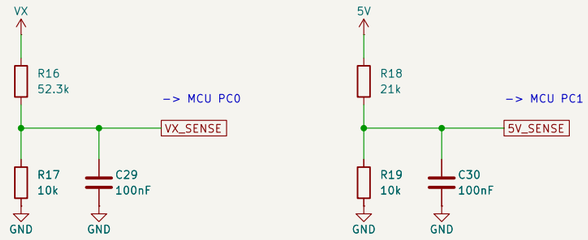


S-CONF 27.4k = setting 9: VSET mode, FB/VSET open = 3.3 V, auto PFM

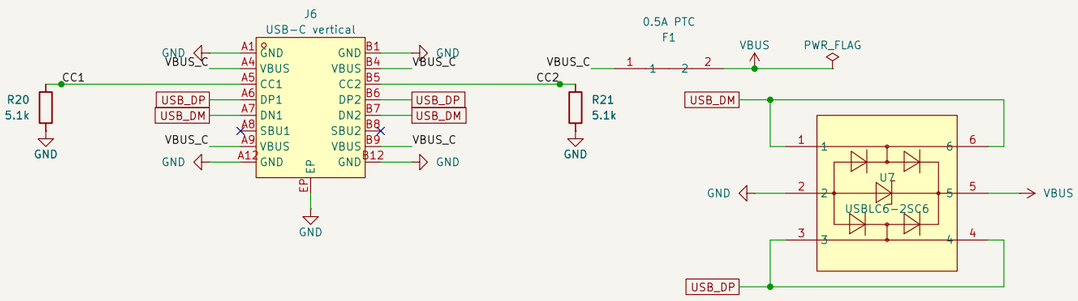
DSM 3V3 PORT FUSE



ADC SENSE VX x0.161 (vbec_divider 160) | 5V x0.323 (vbus_divider 320)



USB-C (VBUS PTC, Rd 5.1k on CC, USBLC6 ESD on D+ / D-)



SBU1 / SBU2 not used; Rd 5.1k on each CC = 5 V sink (UFP)

Parts from nexus-style-fc/tools/design.py (POWER block)
Generated: edit tools/gen_power_schematic.py, then run tools/build.sh

Sheet: /
File: nexus-style-fc-power-module.kicad_sch

Title: Nexus-style FBL — power module

Size: A3	Date: 2026-10-15	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/1